

① a) $F(x) = 0$ IF $x < 0$, OTHERWISE:

$$F(x) = \int_0^x \frac{C}{(u+1)^5} du = C \cdot \left[\left(1 - \frac{1}{4}\right) \cdot (u+1)^{-4} \right]_0^x =$$

$$C \cdot \frac{1}{4} \cdot \left(1 - \frac{1}{(x+1)^4}\right) \xrightarrow{\boxed{C=4}} 1 - \frac{1}{(x+1)^4}$$

LOTUS

$$b) E(X) = E(X+1) - 1 \xrightarrow{\downarrow} \int_0^{\infty} (x+1) \cdot \frac{4}{(x+1)^5} dx - 1 =$$

$$= 4 \cdot \int_0^{\infty} \frac{1}{(x+1)^4} dx - 1 = \frac{4}{3} - 1 = \frac{1}{3}$$

$$E((X+1)^2) \xrightarrow{\text{LOTUS}} \int_0^{\infty} (x+1)^2 \cdot \frac{4}{(x+1)^5} dx = 4 \cdot \frac{1}{2} = 2$$

$$\left(E(X^2) + 2 \cdot E(X) + 1 = E((X+1)^2) \right) \Rightarrow E(X^2) + \frac{2}{3} + 1 =$$

$$= E(X^2) + \frac{5}{3}, \text{ THUS } E(X^2) = \frac{1}{3}$$

$$\text{THUS } \text{Var}(X) = \frac{1}{3} - \left(\frac{1}{3}\right)^2 = \frac{2}{9}$$

c) $Y := \min \{X_1, X_2, X_3\}$ IF $x > 0$ THEN

$$G(x) := P(Y \leq x) = 1 - P(Y > x) =$$

$$1 - P(X_1 \geq x, X_2 \geq x, X_3 \geq x) = \textcircled{0}$$

$$F(x) = 1 - (1 - F(x))^3 = 1 - \frac{1}{(x+1)^{12}} \quad \text{IF } x > 0$$

AND $G(x) = 0$ OTHERWISE.

$$\text{P.D.F. : } g(x) = G'(x) = 12 \cdot \frac{1}{(x+1)^{13}} \cdot \mathbb{I}[x > 0]$$

$$a) P(Y \leq \frac{1}{10}) = 1 - \left(\frac{10}{11}\right)^{12} = 0.6813$$

② Y = NUMBER OF STEW STUDENTS
 $Y \sim \text{BIN}(400, 0.8) \quad E(Y) = 320$

HAPPY IF $Y \leq 320 + n$ AND

$$400 - Y \leq 80 + n \Leftrightarrow Y \geq 320 - n$$

$$0.97 = P(\text{HAPPY}) = P(320 - n \leq Y \leq 320 + n) =$$

$$= P\left(\frac{-n}{\sqrt{400 \cdot 0.8 \cdot 0.2}} \leq \frac{Y - E(Y)}{\sqrt{\text{Var}(Y)}} \leq \frac{n}{\sqrt{400 \cdot 0.8 \cdot 0.2}}\right) \stackrel{\text{DE MOIVRE}}{\approx}$$

$$\approx \Phi\left(\frac{n}{8}\right) - \Phi\left(-\frac{n}{8}\right) = 2 \cdot \Phi\left(\frac{n}{8}\right) - 1 = 0.97$$

$$\Phi\left(\frac{n}{8}\right) = 0.985$$

$$\frac{n}{8} = 2.17$$

$$\boxed{n = 18}$$

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BONUS: SEE GROUP A.

$$X^* = \frac{1}{e^{1/6} - 1}$$